

Group Assignment 1

ADVANCED ECONOMETRICS

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Notes and instructions:

1. This assignment is mandatory.
2. The assignment is to be made in groups of 4 students. You can create your own group in Canvas > People > Assignment Groups, and then self-enroll or join an incomplete team. Please be professional and welcoming to new team members.
3. Only one of you needs to hand in all files.
4. The deadline for delivery of this assignment is on Friday, September 25, at 23:59h. There will be no tolerance period for late deliveries. Deliveries after the assigned deadline imply that you have a final grade of zero for the assignment ($GA1 = 0$).
5. To get the full score for this assignment, the following three things must be done:
 - (a) upload your final report as a PDF file in Canvas Assignments. Name the file **GA1report_2601842_2511351_2661510_2639486.pdf**, where the numbers are replaced by the VU student numbers of the 4 group members. To write your report according to academic standards follow the relevant tips that we have in the questions and also check the example report file under the name '[example_report.pdf](#)' on Canvas > Assignments > Group Assignment 1: Instructions.
 - (b) upload a zip file of your runnable R or Python code in Canvas Assignments. Name the file **GA1code_2601842_2511351_2661510_2639486_language.zip**, where the numbers are replaced by the VU student numbers of the 4 group members (or 3 if your group consists of 3 people) and language is either R or Python. The code file(s) should be clear, well commented, and directly runnable, so that it reads the datafile and obtains the results of all questions and prints them. Your initial comments in the file should hold your names and student numbers.

- (c) upload a pdf of your entire code in Canvas Assignments. Name the file **GA1code_2601842_2511351_2661510_2639486_language.pdf**, where the numbers are replaced by the VU student numbers of the 4 group members (or 3 if your group consists of 3 people) and language is either R or Python. The file should be well readable, with proper indentations and should not contain pictures/photos/screenshots of code snippets.
6. As a standard anti-fraud measure, we will at random select a number of you to explain your code and answers. Any one of you must be able to explain any part of the code. Failure to explain your answers will result in a deduction of credits for this assignment.
 7. For the support for the assignments, carefully read the announcement we put out at the start of the course and consult the discussion boards related to the assignments.

We wish you success!!



Finance: returns and volatilities

You are a data scientist working at (the fictional) Solara Bank. Currently, you are assigned to the financial risk management department, although this is likely to rotate in the future. Solara has large outstanding investments in pharmaceutical companies and wants to understand the potential risks that these investments bring. For example, Solara wants to be able to withstand extreme losses that occur with a chance of at most 5% in each stock.

You will aid Solara's portfolio management and capital allocation by modelling the returns and volatilities of three of its largest pharma stocks in its portfolio. While Solara realizes that the risk of total losses on its pharmaceutical portfolio will be amplified by positive correlation between the share prices of the various stocks, modelling the time-varying volatilities of each individual stock is typically an important first step in a more complex multivariate analysis. Your univariate model estimates will be used to estimate expected future returns, as well as the Value-at-Risk (VaR). You recall from your studies that in economics and finance, it is well known that the conditional volatility of financial returns evolves slowly over time. Financial returns are thus characterized by the presence of 'clusters of volatility'. In addition, a common empirical finding is that large negative returns produce more volatility than large positive returns. This characteristic, first noted by Black (1976), is known as the leverage effect and is something that you surely want to include in your model.

Getting the data

You will have to download the data for the stocks in which Solara is invested yourself. This provides valuable experience with some of the data licenses VU has in store for you that can aid you in conducting high-quality research. For stock price research, Yahoo Finance is not an academically trusted source. Sources that are deemed trustworthy are for instance Datastream, Factset, Compustat-Capital IQ, Center of Security Price Research (CRSP), and others. For this assignment, we take the CRSP data. Be sure to have the UBVU off campus add-in installed if you work off campus: see [this link](#). Go to the [link](#), fill out your VU email address on the left, and do the 'no-robot' test. Then wait for an email (in spam?) from WRDS with your one-day pass for the databases. As data licenses are very expensive, all these safeguards are installed by the data provider.

Clicking the one-day pass link in the email you receive, you enter the WRDS shell. Many high-quality databases are in there, not all of which we have a license for. Some of them may be interesting for your thesis too.

- Find CRSP and click on it.
- Then find ‘Stock/Security files’ and click it.
- Then find ‘Daily Stock File’.
- You now land on a download request form for the data.
- Download the stocks Pfizer (PFE), Johnson & Johnson (JNJ), Merck & Co (MRK) and Apple (AAPL). The latter company (AAPL) is not in Solara’s portfolio, but will be used to check your estimates.
- There are many data types that can be downloaded. You should download the TICKER and the holding period returns (RET). These include dividend payments, and adjust for stock splits and other complications. They are the returns you as an investor would be interested in.
- Download the data from Monday 3 Jan, 2011 to Friday 29 Dec 2023.
- To improve interpretability, multiply the returns by 100 such that 3.5 means 3.5 per cent above average holding period return. As a bonus, this will make your code numerically more stable than working with small numbers like 0.035 meaning 3.5 per cent.

At the very end of this assignment description, you find relevant output to help you on your way.

Univariate volatility modeling

Consider the following model for time-varying volatilities

$$\begin{aligned}
 x_t &= \mu + \lambda \sigma_t^2 + \sigma_t \varepsilon_t \quad \text{for every } t \in \mathbb{Z} \quad \text{where } \{\varepsilon_t\}_{t \in \mathbb{Z}} \sim \text{TID}(v), \\
 \sigma_t^2 &= \omega + (\alpha + \delta \tanh(-\gamma x_{t-1})) \left(\frac{x_{t-1} - \mu - \lambda \sigma_{t-1}^2}{\sigma_{t-1}} \right)^2 + \beta \sigma_{t-1}^2,
 \end{aligned} \tag{1}$$

where $\alpha > |\delta|$, $\beta \geq 0$ and $\gamma > 0$. Here, TID denotes the standard Student’s t distribution with pdf

$$p_\varepsilon(\varepsilon_t) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\Gamma\left(\frac{v}{2}\right) \sqrt{v\pi}} \left(1 + \frac{\varepsilon_t^2}{v}\right)^{-(v+1)/2}.$$

Note that in this volatility model, the volatility may contemporaneously affect the return. In addition, positive returns may have a different impact on the next

period's return than negative returns, the so-called “leverage effect”. Therefore, we will refer to model (1) as a GARCH-M-L model, where M refers to the in-mean effect and L to the leverage effect.

Question 1. Assume that (1) is correctly specified. Show that $\{x_t\}_{t \in \mathbb{Z}}$ is a stationary and ergodic (SE) sequence when

$$\mathbb{E} \log \left(|\gamma| \left(|\lambda \delta| \epsilon_t^2 + \frac{|\delta| |\epsilon_t^3|}{2\sqrt{\omega}} \right) + |\beta| \right) < 0.$$

- Tip: First write out $\sigma_{t+1}^2(\sigma_t^2, \epsilon_t)$, i.e. write the volatility equation as a function of the innovations, and show that $\{\sigma_t^2\}_{t \in \mathbb{Z}}$ is SE. Recall also that

$$\frac{d \tanh(x)}{dx} = 1 - \tanh^2(x).$$

Question 2. Analyze, plot, compare and interpret the news-impact-curves (see Chapter 3, slide 34) of the GARCH-M-L model (1) for different values of $\gamma \in \{0.01, 0.1, 1\}$, and different values of $\delta \in \{0.3, 0.1, 0, -0.3\}$. Create a 2x2 grid of plots where each individual plot contains the news curves for all three gammas corresponding to one specific delta. Make sure your figures for this question satisfy the standards for good scientific reporting (see the tip below Question 3). To plot your curves, use $\mu = 0$, $\lambda = 0$, $\alpha = 0.4$ and set $\sigma_{t-1}^2 = 1$ if it still appears in your news-impact-curve expression. Explain the effect of δ and γ on the model. In particular, what happens as $\delta \rightarrow 0$? What happens if $\gamma \rightarrow \infty$? Do you see any problem with the case $\delta = 0$?

- Tip: In general, the news-impact-curve shows how the shocks affect the conditional volatility. In particular, it shows the values of σ_{t+1}^2 for different values of x_t holding the previous volatility constant, meaning that the function corresponding to the news-impact curve is simply the part of the variance updating equation that is directly affected by x_t (“the news”), while omitting the rest.

Question 3. For this question, use all the data in the sample. Make a table with the descriptive statistics (number of observations, means, medians, standard deviations, skewness, excess kurtosis, minimum, maximum) of your four stocks and briefly discuss it. Also make a figure with four panels, each panel holding the returns for one stock. Also briefly discuss this plot.

- Tip: make sure to obey good academic reporting standards. A good example is given by Figures 3 and 4 in the pdf version of [this paper](#), and for tables in Table 3 of the same paper. If you have trouble accessing the paper, follow [these steps](#) from the University Library to get automatic access with your VU account (a feature that will prove useful at many occasions during your program). Sometimes, the explanatory notes may seem excessive to

you, like in Table 3, but that is not true: you should do everything to make the figures and tables stand-alone, i.e., understandable without reading anything else in the paper. (Note, however, that the explanatory note does not give an interpretation of the data in the figure or table.) Very important: if you make a time series plot, put dates on the axis, and NOT the indices 1, ..., 1000 of the observations.

- To make sure you have the right data, you can check whether you get the following values for AAPL: mean = 0.1071, standard deviation = 1.7832, min=-12.8647, max =11.9808.

Question 4. For each stock, estimate the following three different GARCH models by maximum likelihood **based on the first 2,500 observations**:

- **GARCH**: no mean effect and no leverage ($\lambda = 0; \delta, \gamma = 0$),
- **GARCH-M**: mean effect and no leverage ($\lambda \neq 0; \delta, \gamma = 0$),
- **GARCH-M-L**: mean effect and leverage ($\lambda \neq 0; \delta, \gamma \neq 0$).

To avoid identifiability issues for the GARCH-M-L, consider maximizing the following *penalized* log-likelihood instead:

$$L_T(x_T, \sigma_T(\theta), \theta) - 0.001\gamma^2$$

where $L_T(x_T, \sigma_T(\theta), \theta)$ is the total log-likelihood. Report the (maximum likelihood) estimated parameters of your models in a table. At the end of the table, report the total log-likelihood and the AIC and BIC (see Chapter 10). Make your own (beautiful latex or word) table that satisfies good academic reporting standards (see Table 1 for an example). Use a sensible number of digits as in academic journals. Based on the output in your table, state your preferred model for each of the three stocks. Motivate your choice.

- Tip: to help you on your way, we have provided for you our estimates of the model using stock four (AAPL) in the example Table 1. Some deviation is allowed, given that different optimisers and programming languages yield different optima. However, the likelihood values should be closely replicable (no more deviation than 0.5).
- Tip: we advise to use the 'BFGS' method (is already the default in Python).
- Tip: the conditional density $p_x(x_t | \mathcal{F}_{t-1})$ of x_t , which you need for the construction of the log-likelihood function via the prediction error decomposition, is not simply given by

$$p_\varepsilon\left(\frac{x_t - \mu - c\sigma_t^2}{\sigma_t}\right),$$

where $\mathcal{F}_{t-1} = \{x_{t-1}, x_{t-2}, \dots, x_1\}$ is the information set containing the past observations. Instead, note that for the simpler example $x_t = \sigma_t \varepsilon_t$ with $\varepsilon_t \sim f(\varepsilon_t, \theta)$, the classical theorem for the pdf of a transformation of a random variable tells us that, $x_t = \sigma_t \varepsilon_t \sim f(x_t/\sigma_t, \theta)(1/\sigma_t)$. This means that for this simple example, the log density takes the form $\log f(x_t/\sigma_t, \theta) - \frac{1}{2} \log \sigma_t^2$. Adjust this example to match model (1).

- Tip: since not all parameter restrictions are simply “box constraints”, it may be easiest to guarantee positive variances by simply (heavily) penalizing the log-likelihood in case one of the filtered variances is negative. This way, the optimization can proceed in an unconstrained fashion, which seemed more stable for the data in this assignment.
- Tip: in order to avoid programming errors, you may be tempted to use built-in pdfs and then take the log of those. For example in R, `dnorm(x)` gives the density of a Standard Normal at point x , and `dt(x,lambda)` gives the density of a Student's-t with `lambda` degrees of freedom, at point x . However, this may be a very bad idea in some cases for numerical stability, and in general it is better to implement the expression of the log-likelihood directly. You can use the built-in pdf functions however to numerically check your outcomes.
- Tip: for numerical stability, you do not want to use the log of a gamma function in your log-likelihood expressions. The gamma function rises more than exponentially fast, and the log partly reverts that. This is a recipe for numerical problems. It is better to use the numerically stable log-gamma function, which is built in in all major languages such as R (`lgamma`), Python (`math.lgamma`), and Matlab (`gammaln`) and even Excel (`gammaln`). Just try taking the log of `gamma(10000)`, or computing `loggamma(10000)`.
- Tip: Before optimising, you can always check whether your likelihood function is correct by filling in our optimal estimates, and seeing whether the same optimal likelihood value comes out (this should be exact of course).
- Tip: selecting initial parameter values for the estimation of nonlinear dynamic models is a tricky business! You may want to start the ML parameter optimization with initial parameter values for the three models as

1. $(\mu, \omega, \alpha, \beta, \nu) \approx (0, s_T^2/50, 0.05, 0.9, 10),$
2. $(\mu, \lambda, \omega, \alpha, \beta, \nu) \approx (0, 0, s_T^2/50, 0.05, 0.9, 10),$
3. $(\mu, \lambda, \omega, \alpha, \beta, \delta, \gamma, \nu) \approx (0, 0, s_T^2/50, 0.05, 0.9, 0.01, 0.01, 10).$

This allows for moderately fat-tailed innovations ($\nu = 10$) at the start and no mean effects and (barely any) leverage. We use s_T^2 to denote the sample variance of the returns from $t = 1, \dots, T$ where $T = 2,500$. Such values are typically inspired by earlier work in the academic literature on similar models and similar datasets.

- Tip: there is an issue on how to choose the initial value σ_1^2 . In this assignment, we want you to take the sample variance of the first 50 observations for $\sigma_1^2 = s_{50}^2$. Other choices are possible, but we want you to adopt this specific one for the current case. Note we want you to use $\sigma_1^2 = \frac{1}{50} \sum_{t=1}^{50} (x_t - \bar{x}_{50})^2$ with $\bar{x}_{50} = \frac{1}{50} \sum_{t=1}^{50} x_t$. Note that some packages for the internal `var()` command divide by $T - 1 = 49$ rather than $T = 50$. Check this and correct if necessary in order not to lose any credits, e.g., by checking whether the variance of $\{1, -1\}$ is computed as 1 or 2 (should be 1).
- Tip: Always check whether the optimiser converged! If not, the message saved in the optimiser object should give you an idea of the settings you need to change.

- Tip: note that you have to estimate three separate models per stock. For example, the GARCH-M-L *is not* the same as the estimated GARCH-M with δ and γ manually set to zero. In fact, if you impose no-leverage, all the other parameters like α and β etc. have to be re-estimated under this constraint $\delta, \gamma = 0$.

Table 1: Example

	Explanation								
	AAPL			AAPL			AAPL		
	M1	M2	M3	M1	M2	M3	M1	M2	M3
μ	0.154	0.072	0.108	0.154	0.072	0.108	0.154	0.072	0.108
λ		0.061	0.022		0.061	0.022		0.061	0.022
ω	0.038	0.037	0.012	0.038	0.037	0.012	0.038	0.037	0.012
α	0.090	0.089	0.073	0.090	0.089	0.073	0.090	0.089	0.073
β	0.873	0.875	0.915	0.873	0.875	0.915	0.873	0.875	0.915
δ			0.071			0.071			0.071
γ			0.439			0.439			0.439
ν	4.146	4.138	4.402	4.146	4.138	4.402	4.146	4.138	4.402
Log-lik	-4662	-4662	-4632	-4662	-4662	-4632	-4662	-4662	-4632
AIC	9335	9335	9280	9335	9335	9280	9335	9335	9280
BIC	9364	9370	9327	9364	9370	9327	9364	9370	9327

Question 5. Use the maximum likelihood parameter estimates obtained in the previous question to plot and compare the estimated news-impact-curves and the filtered volatilities obtained from the three alternative models for each of the 3 stocks over the entire sample (so both for the 2,500 in-sample observations, as well as for the remaining out-of-sample observations). Put a vertical line at the *date* where the in-sample observations stop. Summarize all this in a 3×2 panel plot: each row corresponding to a stock (put it in the title of the panel), left-hand plot the three news impact curves, right-hand plot the time series of the filtered volatilities for the GARCH-M-L model. See Figure 1 for an example based on AAPL. Obey good academic reporting standards for the figure. Briefly discuss all plots. As in Question 2, you may use $\sigma_{t-1}^2 = 1$.

Question 6. In this question you compute VaR estimates based on simulation. Use your estimates based on the first 2,500 observations as obtained before. With those estimates, run the filter to compute the volatility up to and including January 4, 2021. On January 4, 2021, you are interested to compute the 1, 5, and 20 day-ahead VaR for the compound returns over these horizons at the levels 1%, 5%, and 10%. Report the results (for each stock you have three models, and for each (stock, model) pair you have 3 horizons and 3 confidence levels, so $3 \times 3 \times 3 \times 3 = 81$ numbers in total. An example is provided in Table 2, where we have filled in our estimated VaRs for AAPL.

Figure 1: Example

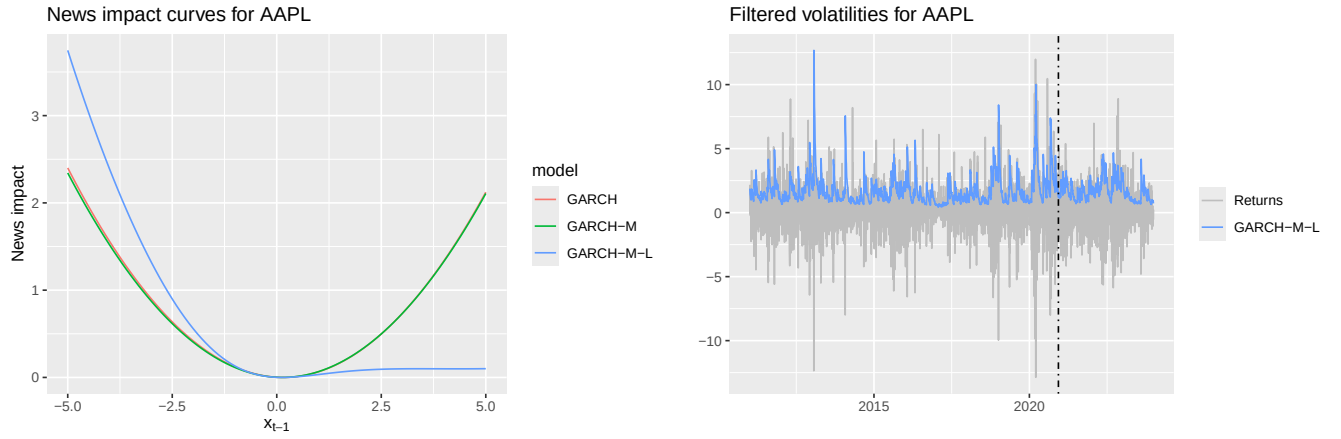


Table 2: Example

Stock	Model	1 step			5 steps			20 steps		
		1%	5%	10%	1%	5%	10%	1%	5%	10%
AAPL	M1	-4.65	-2.58	-1.81	-9.27	-5.40	-3.82	-16.85	-10.03	-6.57
	M2	-4.77	-2.58	-1.83	-8.85	-5.15	-3.71	-15.02	-9.02	-6.32
	M3	-3.91	-2.18	-1.55	-9.04	-4.95	-3.44	-17.07	-9.53	-6.05
AAPL	M1	-4.65	-2.58	-1.81	-9.27	-5.40	-3.82	-16.85	-10.03	-6.57
	M2	-4.77	-2.58	-1.83	-8.85	-5.15	-3.71	-15.02	-9.02	-6.32
	M3	-3.91	-2.18	-1.55	-9.04	-4.95	-3.44	-17.07	-9.53	-6.05
AAPL	M1	-4.65	-2.58	-1.81	-9.27	-5.40	-3.82	-16.85	-10.03	-6.57
	M2	-4.77	-2.58	-1.83	-8.85	-5.15	-3.71	-15.02	-9.02	-6.32
	M3	-3.91	-2.18	-1.55	-9.04	-4.95	-3.44	-17.07	-9.53	-6.05

- Tip: though you are forecasting the VaR, you should not do this in a parametric way. Since we are required to compute multiple step ahead predictions of the VaR, you should instead follow the simulation based approach towards forecasting. As an example for the GARCH model, using $\hat{\sigma}_{t_0}^2(\hat{\theta}, \hat{\sigma}_1^2)$ at t_0 equal to January 4, 2021, compute $\hat{\sigma}_{t_0+1}^2(\hat{\theta}, \hat{\sigma}_1^2)$ using the GARCH equation. Then draw $\varepsilon_{t_0+1}^{(s)}$ from $T(\hat{\sigma})$ and construct $x_{t_0+1}^{(s)} = \hat{\mu} + \hat{\sigma}_{t_0+1}(\hat{\theta}, \hat{\sigma}_1^2) \cdot \varepsilon_{t_0+1}^{(s)}$, where the (s) stands for a simulated HPR (return). Next, compute $\hat{\sigma}_{t_0+2}^{2,(s)}(\hat{\theta}, \hat{\sigma}_1^2)$ using $\hat{\sigma}_{t_0+1}^2(\hat{\theta}, \hat{\sigma}_1^2)$ and $x_{t_0+1}^{(s)}$ and the GARCH equation. Then draw $\varepsilon_{t_0+2}^{(s)}$ and construct $x_{t_0+2}^{(s)} = \hat{\mu} + \hat{\sigma}_{t_0+2}^{(s)}(\hat{\theta}, \hat{\sigma}_1^2) \cdot \varepsilon_{t_0+2}^{(s)}$, and compute $\hat{\sigma}_{t_0+3}^{2,(s)}(\hat{\theta}, \hat{\sigma}_1^2)$ using $\hat{\sigma}_{t_0+2}^{2,(s)}(\hat{\theta}, \hat{\sigma}_1^2)$ and $x_{t_0+2}^{(s)}$ and the GARCH equation, etc. Repeat this procedure until you reach $t_0 + 20$. You can do this for $s = 1, \dots, S$ simulations and then compute the relevant quantiles of your simulated multi-period compound returns to compute the VaR.
- Tip: note that the question is on compound returns. As you have obtained holding period returns x_t from WRDS that you scaled by 100, your compound returns at for instance horizons 1 and 5 are x_{t_0+1} , there t_0 is January 4, 2021, and $100 \cdot [(1 + x_{t_0+1}/100)(1 + x_{t_0+2}/100) \cdots (1 + x_{t_0+5}/100) - 1]$.
- Tip: Beware, the results for AAPL in Table 2 are based on simulations, so you cannot expect to get the same results. You should however be able to get reasonably close. See the output at the end of the assignment. The differences between your outcomes and ours can be substantial for the longer horizons (20) due to compounding slight differences in estimates (even up to plus or minus 2 sometimes). Do not be super-alarmed then (but do triple check your code). In the end, be satisfied if your outcomes land in the rough same ballpark as ours.
- Tip: use (at least) 10,000 simulations in the end (or even more). However, until your code works well, use only 10 simulations or so to prevent having to wait for ages each time. Also note that by simulating for $h = 20$, you have also directly obtained the simulations for $h = 1$ and $h = 5$ if you cleverly store intermediate results.